

Machine Learning Prediction of Customer Churn Status

Session 4: Introduction to Statistical and Machine Learning Methodologies

Victor Wandera Lumumba

Center for Data Analytics and Modelling (CDAM)
Chuka University, Kenya
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Slide 2: Session Outline

- 1 Introduction & Problem Formulation
- 2 Data Understanding & Preprocessing
- 3 Model Development & Algorithms
- 4 Model Evaluation & Metrics
- 5 Empirical Results & Interpretation

Slide 3: Introduction to Machine Learning in Business Analytics

Machine learning transforms historical data into **actionable predictive insights**.

In subscription-driven industries, predicting customer attrition is a **strategic imperative**.

Supervised learning frameworks enable systematic modeling of behavioral and transactional patterns.

This session examines the end-to-end pipeline: data preprocessing → model training → evaluation → business deployment.

Key Insight

Predictive modeling is not an end in itself; it is a decision-support tool for proactive customer retention.

Slide 4: Defining Customer Churn

Operational Definition

Customer churn refers to the termination of a business relationship within a specified observation window. It is particularly consequential in telecommunications, banking, insurance, and streaming services.

Directly impacts **recurring revenue streams** and long-term organizational sustainability.

Early identification enables **proactive retention interventions**.

Formally structured as a **binary classification problem**.

Slide 5: Mathematical Formulation

Let $\mathbf{X} = (x_1, x_2, \dots, x_p)^\top$ denote a feature vector for a given customer. The target variable is defined as:

$$Y = \begin{cases} 1 & \text{if Customer churned} \\ 0 & \text{if Customer retained} \end{cases}$$

The modeling objective is to estimate the conditional probability:

$$\hat{p}(\mathbf{x}) = P(Y = 1 \mid \mathbf{X} = \mathbf{x})$$

A classification threshold τ (typically $\tau = 0.5$) maps probabilities to discrete class labels:

$$\hat{Y} = \mathbb{I}[\hat{p}(\mathbf{x}) \geq \tau]$$

Interpretation

The threshold τ is a **business decision parameter**, not a statistical constant. It should be calibrated based on the relative costs of false positives versus false negatives.

Slide 6: Business Motivation & Strategic Value

Acquiring new customers is typically **5–25× more expensive** than retaining existing ones (Reichheld, 1996).

Churn modeling shifts strategy from **reactive** to **proactive retention**.

Predictive scores enable resource allocation toward high-value, at-risk segments.

Ultimate goal: maximize **Customer Lifetime Value (CLV)** and organizational profitability.

Strategic Implication

Model performance metrics must align with business objectives. High statistical accuracy does not guarantee business utility if the model fails to identify actionable churners.

Slide 7: Feature Categories: Demographic Variables

Include age, gender, income level, geographic location, senior citizen status.

Primarily used for customer segmentation and baseline profiling.

Often exhibit **limited standalone predictive power** but provide essential context when combined with behavioral signals.

Interpretation

Demographic features establish *who* the customer is, but rarely explain *why* churn occurs. Their utility increases when interacted with engagement or contractual variables (e.g., “senior citizen $\times$ month-to-month contract”).

Slide 8: Feature Categories: Behavioral Variables

Purchase frequency, average transaction value, service interaction count.

Product usage frequency, complaint history, days since last activity.

Capture **dynamic engagement patterns** and service satisfaction.

Typically the **strongest predictors** of imminent churn.

Interpretation

Behavioral metrics reflect *real-time engagement decay*. A sharp decline in usage or spike in support tickets often precedes contractual termination by 30–60 days, providing a critical intervention window.

Slide 9: Feature Categories: Contractual & Subscription Variables

Tenure (duration of relationship), contract type (month-to-month, annual, biennial).

Payment method, subscription tier, renewal history.

Tenure is frequently among the **most robust predictors**: shorter tenure correlates with higher churn risk.

Long-term contracts inherently reduce attrition probability through structural commitment.

Interpretation

Contractual features capture *structural lock-in* and *historical commitment*, serving as strong baseline indicators of retention likelihood. Month-to-month contracts exhibit 3–4× higher churn rates than two-year agreements in telecom datasets.

Slide 10: Data Preprocessing: Missing Value Imputation

Real-world datasets invariably contain missing entries due to system errors or non-response.

Numerical variables: Impute using mean or median, depending on distribution skewness.

Categorical variables: Impute using mode or introduce a “Missing” category.

TotalCharges often requires type coercion (`object` → `numeric`) before imputation.

Interpretation

Improper imputation can introduce bias. For skewed financial metrics like TotalCharges, **median imputation** is preferred over mean to avoid inflating variance and distorting model coefficients.

Slide 11: Data Preprocessing: Encoding & Feature Scaling

Categorical Encoding:

- Label encoding for ordinal variables (e.g., contract length).

- One-hot encoding for nominal variables (`drop_first=True` to avoid multicollinearity).

Feature Scaling: Standardization $z = \frac{x - \mu}{\sigma}$ is critical for:

- Distance-based algorithms (KNN, SVM with RBF kernel)

- Gradient-based optimization (Logistic Regression, Neural Networks)

Tree-based models (Random Forest, XGBoost) are invariant to monotonic transformations.

Slide 12: Handling Class Imbalance

Churn datasets are typically imbalanced ($N_{\text{retained}} \gg N_{\text{churned}}$). In our dataset: $\approx 73\%$ “No”, $\approx 27\%$ “Yes”.

Class Weighting: Penalize misclassification of minority class during training (`class_weight='balanced'`).

Over-sampling: SMOTE generates synthetic churner instances in feature space.

Under-sampling: Reduces majority class size; risks information loss.

Evaluation must prioritize **recall** and **F1-score** over raw accuracy.

Critical Warning

Ignoring imbalance yields models that predict “No Churn” universally, achieving high accuracy ($> 70\%$) but **zero business utility**. Always report class-wise metrics.

Model Development Pipeline

Baseline model: Logistic Regression

Advanced models: Random Forest, XGBoost

Data split: 70/30 or 80/20

Validation: k-fold cross-validation

Training Set -> Model -> Prediction -> Evaluation -> Deployment

Best Practice

Keep a separate test set and only use it for final evaluation.

Slide 14: Logistic Regression: Baseline Modeling

Models the log-odds of churn as a linear combination of predictors:

$$\log \left(\frac{P(Y = 1 \mid \mathbf{X})}{1 - P(Y = 1 \mid \mathbf{X})} \right) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

Highly interpretable; coefficients indicate directional influence and marginal effects.

Serves as a benchmark for complexity vs. performance trade-offs.

Assumes linear decision boundary in log-odds space.

Interpretation of Coefficients

For a binary predictor x_j , $\exp(\beta_j)$ represents the **odds ratio**: the multiplicative change in churn odds associated with a one-unit increase in x_j , holding other variables constant.

Slide 15: Ensemble Methods: Random Forest & Gradient Boosting

Random Forest: Bagging-based ensemble; reduces variance via averaging multiple decorrelated trees. Robust to noise, non-linearities, and feature interactions.

Gradient Boosting (XGBoost/LightGBM): Sequential learning; each tree corrects residuals of predecessors. Optimizes differentiable loss functions via gradient descent.

Typically outperforms single models in churn prediction benchmarks (Chen et al., 2021).

Interpretation

Ensembles capture complex interaction effects (e.g., *low tenure* $\times$ *high monthly charges* $\times$ *fiber optic service*) that linear models miss. However, they sacrifice some interpretability for predictive gain.

Slide 16: K-Nearest Neighbors (KNN)

Instance-based, non-parametric algorithm.

Classifies a query point based on majority vote among k nearest neighbors in feature space.

Sensitive to feature scaling and high-dimensional sparsity (curse of dimensionality).

$k = 10$ often balances bias-variance trade-off in moderate-sized datasets.

Interpretation

KNN assumes **locally homogeneous churn behavior**: customers with similar feature profiles exhibit similar churn propensity. Performance degrades if churn is driven by global thresholds rather than local clusters.

Slide 17: Support Vector Machines (SVM)

Maximizes margin between classes in transformed feature space.

Kernels (linear, RBF, polynomial) enable non-linear separation.

Requires explicit probability calibration (`probability=True`) for ROC/AUC computation.

Computationally intensive for large datasets; sensitive to imbalance without class weighting.

Critical Limitation

Default SVM exhibits severe majority-class bias in imbalanced settings. Without `class_weight` or SMOTE preprocessing, it may predict **all instances as the majority class**, yielding high accuracy but zero recall for churners.

Slide 18: Evaluation Metrics: Precision & Recall

Precision: $\frac{TP}{TP + FP}$

Proportion of predicted churners who actually churned. High precision minimizes false alarms and wasted retention resources.

Recall (Sensitivity): $\frac{TP}{TP + FN}$

Proportion of actual churners correctly identified. Critical for retention strategy.

Business Interpretation

In churn modeling, **recall is prioritized** over precision. Missing a churner (FN) represents irreversible revenue loss, whereas false positives (FP) incur manageable retention costs (discounts, outreach). The optimal trade-off depends on the organization's cost structure.

Slide 19: F1-Score, Support, & Averaging Strategies

F1-Score: Harmonic mean of precision and recall:

$$F_1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Useful when seeking a single metric that balances both concerns.

Support: Actual number of instances per class in the test set.

Macro Average: Unweighted mean across classes; treats minority/majority equally (sensitive to imbalance).

Weighted Average: Mean weighted by support; reflects overall dataset performance.

Recommendation

For imbalanced churn data, report **both** macro and weighted averages, and always inspect class-wise metrics. Do not rely solely on aggregate scores.

Slide 20: ROC Curve & AUC

ROC Curve: Plots True Positive Rate (Recall) against False Positive Rate across varying classification thresholds.

AUC: Area Under the Curve; threshold-independent measure of separability.

Interpretation:

AUC = 0.5 → random guessing

AUC = 0.7–0.8 → acceptable discrimination

AUC = 0.8–0.9 → excellent discrimination

AUC = 1.0 → perfect classifier

Particularly valuable when class distribution or business costs shift over time.

Interpretation

A high AUC indicates the model **ranks** churners higher than non-churners consistently, regardless of the chosen decision threshold. It measures discriminative ability, not calibration.

Slide 21: Confusion Matrix Analysis

	Pred: No	Pred: Yes
Actual: No	TN	FP
Actual: Yes	FN	TP

Visualizes classification performance across both classes.

Reveals systematic biases (e.g., majority-class prediction).

Essential for diagnosing recall-precision trade-offs.

Enables cost-benefit analysis: assign monetary values to FP and FN errors.

Slide 22: KNN Model Results ($k = 10$)

Overall Accuracy: 77.19%

Non-Churner (No): Precision = 0.79, Recall = 0.93, F1 = 0.86

Churner (Yes): Precision = 0.66, Recall = 0.35, F1 = 0.46

Macro Avg F1: 0.66 | **Weighted Avg F1:** 0.75

ROC-AUC: ≈ 0.72 (moderate discriminative ability)

Critical Interpretation

KNN captures majority class patterns well but misses $\sim 65\%$ of actual churners (Recall = 0.35). The low recall indicates **insufficient sensitivity** to churn signals, likely due to: (1) class imbalance, (2) distance-based smoothing diluting minority-class signals, and (3) lack of explicit class weighting. For retention purposes, this model would fail to trigger interventions for most at-risk customers.

Slide 23: SVM Model Results (Default Configuration)

Overall Accuracy: 73.53%

Non-Churner (No): Precision = 0.74, Recall = 1.00, F1 = 0.85

Churner (Yes): Precision = 0.00, Recall = 0.00, F1 = 0.00

Macro Avg F1: 0.42 | **Weighted Avg F1:** 0.62

Confusion Matrix: All 373 actual churners predicted as “No”

Critical Interpretation

The default SVM exhibits **severe majority-class bias**, predicting all instances as “No Churn”. Despite moderate accuracy (73.5%), it is **operationally useless** for retention: zero churners identified. This failure stems from: (1) extreme sensitivity to class imbalance without `class_weight`, (2) RBF kernel overfitting to majority-class density, and (3) default threshold ($\tau = 0.5$) inappropriate for imbalanced data. Remediation requires explicit class weighting, SMOTE, or threshold optimization.

Feature Importance & Explainability

Tree-based models show feature importance automatically.

Permutation importance checks impact by shuffling features.

Key churn drivers in our dataset:

- TotalCharges

- MonthlyCharges

- tenure

SHAP values explain individual predictions.

Business Insight

High monthly charges combined with short tenure strongly increase churn risk.

Risk Segmentation & Retention Strategy

High Risk: $P \geq 0.7 \rightarrow$ urgent retention actions

Medium Risk: $0.4 \leq P < 0.7 \rightarrow$ engagement campaigns

Low Risk: $P < 0.4 \rightarrow$ monitor and upsell

Helps optimize retention budget

Implementation Tip

Use historical data to set realistic probability thresholds.

Threshold Optimization

Default threshold (0.5) is not always optimal.

False negatives are more costly than false positives in churn.

Lower threshold $\rightarrow$ higher recall (catch more churners).

Optimize using cost or profit analysis.

Key Insight

Threshold choice should reflect business cost, not just model accuracy.

Advanced Methodological Extensions

Survival analysis: predicts time until churn.

Time series models: capture behavior over time.

Ensemble stacking: combine multiple models.

Real-time scoring via APIs.

Research Direction

Use text data (calls, chats) to improve churn prediction.

Implementation Guidelines

Focus on data quality over complex models.

Use AUC, recall, and F1 instead of accuracy alone.

Apply explainability tools such as SHAP and LIME.

Monitor model drift regularly.

Align outputs with business processes.

Governance: Ensure compliance with data privacy regulations and document model usage.

Conclusion

Churn prediction is a supervised classification problem.

Data imbalance must be properly handled.

Ensemble models usually perform best.

Business value matters more than accuracy alone.

Final Message

A model is only useful if it drives better business decisions.

Thank You

Victor Wandera Lumumba, Ph.D.
Center for Data Analytics and Modelling (CDAM)
Chuka University, Kenya

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Questions, Comments, and Discussion